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Ans 1. Newtons 2nd law - rate of change of momentum of a body is proportional to force applied on it n happens in direction of force. $F=ma$

Ans 2. speed is scalar only magnitude eg 40km/hr, velocity is vector has magnitude n direction eg 40km/hr north

Ans 4. SI unit of current is Ampere. 1 ampere = current when 1 coulomb charge flows in 1 sec thru conductor

Ans 6. $u=0$ $a=2$ $t=5$
 $v=u+at=0+2(5)=10\text{m/s}$
 $s=ut+\frac{1}{2}at^2=0+\frac{1}{2}(2)(25)=25\text{m}$

Ans 8. ohms law $V=IR$ at const temp current prop to voltage. $V=IR=2 \times 10=20\text{volts}$

Ans 11 (a). $v=u+at$. (1) avg vel $=(u+v)/2$ $s=\text{avg vel} \times t = [(u+v)/2]t$ sub v from (1) $s=[(u+u+at)/2]t=ut+\frac{1}{2}at^2$

Ans 9. 3 states - solid liquid gas. solid particles tight packed. liquid particles move but close. gas particles far apart move free

Ans 7. mitosis - 2 identical daughter cells for growth/repair. meiosis - 4 cells half chromosome for reproduction

Ans 12. 1 rainwater harvesting - collect rain for reuse. 2 fix leaky pipes/taps stop wastage

physics formulas for revision - $F=ma$ $v=u+at$ $s=ut+\frac{1}{2}at^2$
 $v^2=u^2+2as$ $V=IL$ $P=VI$